

# Arduino Based Gesture-Controlled Robot

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**P**resented here is an Arduino based gesture-controlled robot, which is controlled through an accelerometer module using wireless radio frequency (RF) communication.

## Circuit and working

In this project, we have used two Arduino UNO boards, one at the transmitter side as shown in Fig. 1 and the other at the receiver side as shown in Fig. 2. The transmitter circuit is attached to the hand and the receiver



circuit is attached to mechanical chassis of the robot.

Hand gestures, as shown in Fig. 3, can control the robot using an accelerometer such as ADXL345 module. By changing the axis of ADXL345, we get corresponding changes in X-axis, Y-axis and Z-axis coordinates. These are sent to Arduino Board2 and, accordingly, control

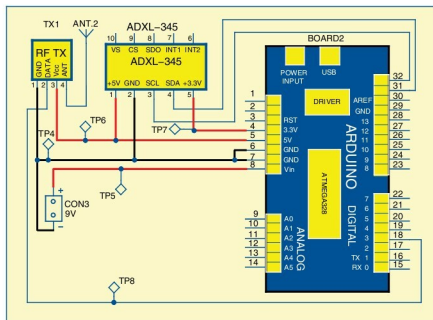


Fig. 1: Transmitter-side circuit

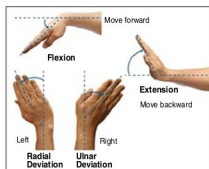


Fig. 3: Various hand gestures

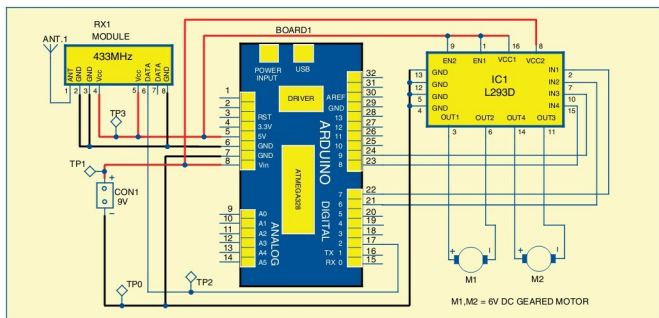


Fig. 2: Receiver-side circuit